

Scope Management Plan

1. Introduction

1.1 Overview of the Project

The **Mumbai–Ahmedabad High-Speed Rail Corridor** is a flagship transportation initiative aimed at enhancing rail efficiency in India. The project focuses on **developing a scheduling solution** that leverages **AI and ML algorithms** to optimize train timetables, reduce delays, and enhance passenger experience. The **National High-Speed Rail Corporation Limited (NHSRCL)** is the client, and **viaTeams** is responsible for proposal development.

1.2 Purpose of the Scope Management Plan

The Scope Management Plan defines how the project scope will be **defined, validated, and controlled** throughout its lifecycle. It ensures that project deliverables align with **NHSRCL's objectives**, minimizing scope creep and ensuring successful delivery.

2. Scope Management Approach

2.1 Scope Management Process

The scope will be managed through the following key steps:

- **Requirements gathering:** Collaboration with NHSRCL to define system needs.
- **Scope definition:** Creation of a **detailed project scope statement**.
- **Scope validation:** NHSRCL will formally approve project milestones.
- **Scope control:** Change management procedures to handle scope modifications.

2.2 Roles and Responsibilities

- **Project Manager (Roger Candoo):** Ensures scope is aligned with NHSRCL expectations and manages scope changes.
 - **Stakeholder Representative (NHSRCL Officials):** Validates project requirements and deliverables.
 - **Technical Lead:** Ensures feasibility of AI/ML-based scheduling system.
 - **Quality Assurance Lead:** Verifies that the scope adheres to regulatory and performance standards.
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3. Scope Definition

3.1 Process for Scope Definition

- **Stakeholder Workshops:** Gathering requirements from NHRCL and regulatory bodies.
- **Business Analysis:** Defining **functional and non-functional requirements**.
- **Technical Feasibility Study:** Evaluating available technologies for AI-based scheduling.

3.2 Techniques & Tools

- **Requirements Documentation:** Clearly defining needs in collaboration with NHRCL.
 - **Prototyping:** Developing initial scheduling models for validation.
 - **Stakeholder Interviews:** Ensuring all operational requirements are met.
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4. Work Breakdown Structure (WBS)

4.1 Process for Creating the WBS

The WBS is developed using a **top-down decomposition** approach, aligned with NHRCL's operational needs.

4.2 Guidelines for WBS Development

- Deliverables-based decomposition ensuring clarity of scope.
- Hierarchical structure with measurable outputs.

4.3 WBS Dictionary

- **Level 1: Project Initiation** (Stakeholder engagement, requirements analysis).
 - **Level 2: Analysis Phase** (Data collection, demand analysis, technology assessment).
 - **Level 3: Timetable Development** (Drafting, simulation, optimization).
 - **Level 4: Proposal Preparation** (Risk analysis, documentation, cost estimates).
 - **Level 5: Submission & Follow-up** (Presentation, NHRCL feedback incorporation).
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5. Scope Verification

5.1 Acceptance Criteria

- All deliverables must align with **NHRCL's requirements**.
- Timetable optimization models must achieve at least **95% accuracy**.
- Compliance with **Indian Railways and UIC regulations**.

5.2 Process for Formal Acceptance

- NHSRCL will review **each project phase**.
 - Final approval required from **NHSRCL Steering Committee**.
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6. Scope Control

6.1 Change Management Process

Scope changes will follow an **Integrated Change Control process**, ensuring:

- **Documented change requests**.
- **Impact analysis** (cost, time, and quality implications).
- **Formal approval by NHSRCL** before implementation.

6.2 Handling Scope Creep

To prevent scope creep:

- All **new requirements** must be validated through a **change request process**.
- Regular **stakeholder reviews** will ensure scope alignment.

6.3 Integrated Change Control

Changes affecting the project scope will require:

- **Approval from the Project Sponsor and NHSRCL**.
 - Updates to project documents (WBS, schedule, cost estimates).
 - A **risk assessment** of potential impacts.
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7. Scope Baseline

The scope baseline includes:

- **Scope Statement**: Defines project objectives and deliverables.
- **WBS**: Breakdown of deliverables.
- **WBS Dictionary**: Detailed description of each WBS element.

The baseline will be maintained through a **formal approval process** and updated **only via approved change requests**.

8. Scope Management Roles and Responsibilities

8.1 Key Stakeholders

- **NHSRCL Steering Committee:** Approves scope and key deliverables.
- **Project Sponsor (Nicole Schelter, CEO of viaTeams):** Ensures project alignment with strategic goals.
- **Project Manager (Roger Candoo):** Manages day-to-day scope execution.
- **Technical Team:** Implements scheduling solution.
- **Quality Assurance Team:** Ensures compliance and performance standards.

8.2 Specific Responsibilities

Role	Responsibility
Project Manager	Manages scope execution and controls changes
NHSRCL Representatives	Validate deliverables and ensure alignment
Technical Team	Develops scheduling solution
Quality Assurance Team	Conducts system testing and verification

9. Scope Management Tools and Techniques

9.1 Tools for Scope Management

- **Requirements Documentation** (to ensure clear definition).
- **Work Breakdown Structure (WBS)** (to define project scope).
- **Traceability Matrix** (to track requirements fulfillment).
- **Scope Change Control System** (to manage modifications).

9.2 Examples of Tools Used

- **JIRA** (for issue and change tracking).
 - **Prototyping Tools** (for early-stage validation).
 - **Simulation Software** (for timetable optimization).
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10. Scope Management Performance Measurement

10.1 Key Performance Indicators (KPIs)

- **Scope Adherence:** % of deliverables completed within scope.
- **Change Control Effectiveness:** % of changes approved vs. requested.

- **Requirement Fulfillment:** % of requirements met.

10.2 Scope Reporting Process

- **Weekly progress reports** to NHSRCL.
 - **Monthly scope review meetings.**
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11. Scope Management Communication

11.1 Communication Plan

Scope-related updates will be communicated via:

- **Weekly project status reports.**
- **Monthly NHSRCL review meetings.**
- **Change management meetings** (as needed).

11.2 Communication Methods

Stakeholder	Communication Method	Frequency
NHSRCL Steering Committee	Formal reports & meetings	Monthly
Project Sponsor	Status updates	Bi-weekly
Development Team	Daily stand-ups	Daily
QA Team	Review meetings	Weekly
